

Ch12: Ethical Aspects of Emerging and Converging Technologies

P3: Autonomous Machines



50

3. Autonomous Machines (AMs)

- Autonomous machines (AMs) are yet another example of an emerging technology that can have a significant ethical impact.
- AMs include any computerized system/agent/robot that is capable of acting and making decisions independently of human oversight.
- An AM also can interact with and adapt to (changes in) its environment and it can learn (as it functions).

52



54

Module Objectives

Upon completion of this module the students should be able to:

- Define “autonomous machines” and discuss the ethical challenges they pose

51

AMs (Continued)

- The expression “autonomous machine” includes three conceptually distinct, but sometimes overlapping, *autonomous* technologies:
 - 1) (autonomous) artificial agents,
 - 2) autonomous systems,
 - 3) (autonomous as opposed to “tele”) robots.
- The key attribute that links together these otherwise distinct (software) programs, systems, and entities is their ability to act *autonomously*, or at least act independently of human intervention.

53



55



AMs: Some Examples and Applications

- driverless transport systems (in commerce);
- unmanned vehicles in military/defense applications (e.g., “drones”);
- robots on the battlefield;
- autonomous robotic surgery devices;
- personal care support systems.

56



Can an AM Be an (Artificial) Moral Agent?

- Luciano Floridi (2011) believes that AMs can be moral agents because they
 - (a) are “sources of moral action”;
 - (b) can cause moral harm or moral good.
- In Chapter 11, we saw that Floridi distinguished between “moral patients” (as receivers of moral action) and moral agents (as sources of moral action).
- All information entities, in Floridi’s view, deserve consideration (minimally at least) as moral patients, even if they are unable to qualify as moral agents.

58



AMs as Moral Agents: Moor’s Model

- James Moor (2006) takes a different tack in analyzing this question by focusing on various kinds of “moral impacts” that AMs can have.
- First, he notes that computers can be viewed as *normative* (non-moral) agents – independently of the question whether they are also moral agents – because of the “normative impacts” their actions have (irrespective of any moral impacts).

60



AMs (Continued): Some Examples and Applications

- Patrick Lin (2012) identifies a wide range of sectors in which AMs (or what he calls “robots”) now operate., six of which include:
 - labor and service,
 - military and security,
 - research and education,
 - entertainment,
 - medical and healthcare,
 - personal care and companionship.

57



AMs as Moral Agents (Continued)

- Floridi also believes that autonomous AMs would qualify as moral agents because of their (moral) efficacy.
- Deborah Johnson (2006), who also believes that AMs have moral efficacy, argues that AMS qualify only as “moral entities” and not moral agents because AMs lack freedom.
- Himma (2009) argues that because these entities lack consciousness and intentionality, they cannot satisfy the conditions for moral agency.

59



Moor’s Model (Continued)

- Moor also points out that because computers are designed for specific purposes, they can be evaluated in terms of how well, or how poorly, they perform in accomplishing the tasks they are programmed to carry out.
- For example, he considers the case of a computer program designed to play chess (such as Deep Blue) that can be evaluated *normatively* (independent of ethics).

61

Moor's Model (Continued)

- Moor notes that some normative impacts made possible by computers can also be moral or ethical in nature.
- He argues that the consequences, and potential consequences, of (what he calls) "ethical agents" can be analyzed in terms of four levels:
 1. Ethical Impact Agents,
 2. Implicit Ethical Agents,
 3. Explicit Ethical Agents,
 4. Full Ethical Agents.

62

Moor's Model (Continued)

- In Moor's scheme:
- *ethical-impact-agents* (i.e., the weakest sense of moral agent) will have (at least some) ethical consequences to their acts;
- *implicit-ethical-agents* have some ethical considerations built into their design and "will employ some automatic ethical actions for fixed situations";
- *explicit-ethical-agents* will have, or at least act as if they have, "more general principles or rules of ethical conduct that are adjusted and interpreted to fit various kinds of situations";
- *full-ethical agents* "can make ethical judgments about a wide variety of situations" and in many cases can "provide some justification for them."

63

Moor's Model (Continued)

- Moor provides some examples of the first two categories:
- 1. An *ethical-impact agent* can include a "robotic camel jockey" (a technology used in Qatar to replace young boys as jockeys, and thus freeing those boys from slavery in the human trafficking business).
- 2. *Implicit ethical agents* include an airplane's automatic pilot system and an ATM – both have built-in programming designed to prevent harm from happening to the aircraft, and to prevent ATM customers from being short-changed in financial transactions.
- 3. *Explicit ethical agents* would be able to calculate the best ethical action to take in a specific situation and would be able to make decisions when presented with ethical dilemmas.
- 4. *Full-ethical agents* have the kind of ethical features that we usually attribute to ethical agents like us (i.e., what Moor describes as "normal human adults"), including consciousness and free will.

64

Moor's Model (Continued)

- Moor does not claim that either explicit- or full-ethical (artificial) agents exist or that they will be available anytime in the near term.
- But his distinctions are very helpful, as we try to understand various levels of moral agency that potentially affect AAs.
- Even if AMs may never qualify as full moral agents, Wallach and Allen (2009) believe that they can have "functional morality," based on two key dimensions:
 - i. autonomy,
 - ii. sensitivity to ethical values.

65

Wallach and Allen's Criteria for "Functional Morality" for AMs

- Wallach and Allen also note that *we do not yet have systems with both high autonomy and high sensitivity*.
- They point out that an *autopilot is an example of a system that has significant autonomy (in a limited domain) but little sensitivity to ethical values*.
- Wallach and Allen also note that ethical-decision support systems (such as those used in the medical field to assist doctors) provide decision makers with access to morally relevant information (and thus suggest a high level of sensitivity to moral values), but these systems have virtually no autonomy.

66

Functional Morality (Continued)

- Wallach and Allen argue that it is not necessary that AMs be moral agents in the sense that humans are.
- They believe that all we need to do is to design machines to act "as if" they are moral agents and thus "function" as such.

67

In What Sense of "Autonomy" are AMs Autonomous?

- Onora O'Neill (2002) describes autonomy in connection with one's "capacity to act independently."
- Wha-Chul Son (2015) points out that that AMs (or "autonomous technologies") can undermine "human autonomy" in both indirect and subtle ways.
- The Royal Academy's 2009 influential report suggests that because AMs (autonomous systems) are "adaptive," they exhibit some degree of "independence."
- Floridi (2008) notes that an "adaptive" AM has a certain degree of "independence from its environment."

68

"Functional Autonomy" for AMs

- In so far as AMs appear to be capable of acting independently, or behave "as if" they are acting independently, it would seem that we can attribute at least some degree of *autonomy* to them.
- Whether AMs will ever be capable of having full autonomy, in the sense that humans can, is still debatable.
- But an AM that can act independently in the sense described above can have "functional autonomy" and thus can qualify as a "functionally autonomous AM."

69

Trust and Authenticity in the Context of AMs

- What is trust in the context of AMs, and what does a trust relationship involving humans and AMs entail?
- For example, can we trust AMs to always act in our best interests, especially AMs designed in such a way that they cannot be shut down by human operators?
- We limit our discussion to two basic questions:
 - I. What would it mean for a human to trust an AM?
 - II. Why is that question important?
- First, we need to define what is meant by trust in general.

70

What is *Trust*?

- A typical dictionary, such as the *American Heritage College Dictionary* (4th ed. 2002), defines trust as "firm reliance on the integrity, ability, or character of a person or thing."
- Definitions of trust that focus mainly on *reliance* do not help us to understand the nature of ethical trust.
- For example, I *rely* on my automobile engine to start today but I do not "trust" it to do so.
- Conversely, I trust some people but I cannot always rely upon them for specific tasks.

71

Trust (Continued)

- Because I am unable to have a trust relationship with a conventional machine such as an automobile, does it also follow that I also cannot have one with an AM?
- Or, does an AM's ability to exhibit some level of autonomy – even if only functional autonomy – make a difference?
- Consider that I am able to trust a human because the person in whom I place my trust not only can disappoint me (or let me down) but can also betray me.
- For example, that person, as an autonomous agent, can freely elect to breach the trust I placed in her.

72

Trust (Continued)

- Some argue that trust also has an emotive (or "affective") aspect, and that this may be especially important in understanding trust in the context of AMs.
- For example, Sherry Turkle (2011) raises some concerns about the role of *feelings* or *emotions* in human-machine trust relationships.
- Turkle also worries about what can happen when machines appear to us "as if" they have feelings.
- Turkle describes a phenomenon called the "Eliza effect," which was initially associated with a response that some users had to an interactive software program called "Eliza" (designed by Joseph Weizenbaum at MIT in the 1960s).

73



Trust (Continued)

- Turkle notes that the Eliza program, which was designed to use language conversationally (and possibly pass the Turing test), solicited trust on the part of users.
 - The Eliza program did this, even though it was designed in a way that tricks users.
 - Although Eliza was only a (“disembodied”) software program, Turkle suggests that it could nevertheless be viewed as a “relational entity,” or what she calls a “relational artifact,” because of the way people responded to, and confided in, it.
 - In this sense, she notes that Eliza seemed to have a strong emotional impact on the students who interacted with it.
 - But Turkle also notes that while Eliza “elicited trust” on the part of these students, it understood nothing about them.
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74



Trust and “Attachment” in AMs

- Turkle worries that when a machine (as a relational artifact) appears to be interested in people, it can “push our Darwinian buttons...which causes people to respond *as if* they were in a relationship.”
 - This is especially apparent with physical AMs that are capable of facial expressions, such as (the robot) *Kismet* (developed in MIT’s AI Lab).
 - Turkle suggests that because AMs can be designed in ways that make people feel as if a machine (like *Kismet*) cares about them, people can develop feelings of trust in, and attachment to, that machine.
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75



Trust and “Authenticity”

- Turkle notes that Cynthia Breazeal, one of *Kismet*’s designers who had also developed a “maternal connection” with this AM while she was a student at MIT, had a difficult time separating from *Kismet* when she left that institution.
 - In Turkle’s view, this factor raises questions of both trust *and* authenticity
 - She worries that, unlike in the past, humans must now be able to distinguish between authentic and simulated relationships.
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76